La Medicina del Lavoro

Original article

Med. Lav. 2026; 117 (4): 18572
DOI: 10.23749/mdl.2026.18572

Women's Work and Its Effects on Personal Well-being and Satisfaction With Life: A Cross-Sectional Comparative Study

Hanan Mohammed[1], Asmaa Awaad[2,*], Abdel-Hady El-Gilany[3], Samar Abdelraouf[1], Eman El-Refaay[1]

[1]Community Health Nursing, Faculty of Nursing, Mansoura University, Mansoura, Egypt
[2]Industrial Medicine and Occupational Health, Public Health & Community Medicine Department, Faculty of Medicine, Mansoura University, Mansoura, Egypt
[3]Public Health and Preventive Medicine, Public Health & Community Medicine Department, Faculty of Medicine, Mansoura University, Mansoura, Egypt

Keywords: Life Satisfaction; Subjective Wellbeing; Work-Life Balance; Working women

Abstract

Background: *Women's participation in the workforce has increased substantially, raising important questions about how employment is associated with their subjective well-being and satisfaction with life. This study aimed to assess differences in personal well-being and life satisfaction between working and non-working women and to identify their significant socio-demographic and occupational predictors.* **Methods:** *A cross-sectional comparative study was conducted among 402 women (192 working, 210 non-working) attending outpatient clinics at Mansoura University and primary healthcare centers. Data were collected using an interviewer-administered questionnaire covering sociodemographic factors, occupational characteristics, the Personal Well-being Index–Adult (PWI-A), and the Life Satisfaction Scale(LSS). Bivariate and binary logistic regression analyses identified the independent predictors.* **Results:** *Working women reported significantly higher well-being (PWI-A: 72.06 ± 13.03) and life satisfaction (LSS: 21.52 ± 5.76) than non-working women (PWI-A: 67.42 ± 10.14; LSS: 17.55 ± 5.91). University education (AOR = 2.81) and sufficient income (AOR = 2.94) were strong predictors of higher wellbeing. Subjective well-being (AOR = 21.67) was a significant predictor of life satisfaction. Among working women, professional (AOR = 4.76) and administrative/technical occupations (AOR = 7.40), and absence of shift work (AOR = 6.53) were associated with better wellbeing. Shorter working hours (≤40 h/week; AOR = 14.34) were linked to higher life satisfaction.* **Conclusion:** *In Mansoura, working women exhibit higher subjective well-being and life satisfaction than non-working women. These outcomes were associated with education, income, and occupational conditions. Improving work conditions may contribute to better well-being among working women.*

1. Introduction

Women constitute a substantial proportion of the population in all countries, and their social, economic, and professional participation is increasingly recognized as a key determinant of national development. In recent decades, the rising engagement of women in paid employment has emerged as

Received 16.02.2026 – Accepted 15.06.2026
*Corresponding Author: Asmaa Awaad; E-mail: asmaa_awaad@mans.edu.eg

an important demographic and socioeconomic shift. This trend warrants focused investigation, particularly in relation to women's subjective well-being, as their psychological and emotional states may be associated with productivity, organizational performance, and broader societal outcomes [1].

Well-being is a multidimensional concept that extends beyond the absence of illness and encompasses both emotional experiences and effective functioning in daily life. It includes the presence of positive affective states, such as happiness and contentment, alongside the ability to realize personal capacities, exercise autonomy, pursue meaningful goals, and maintain supportive interpersonal relationships [2].

Subjective well-being (SWB) refers to individuals' personal evaluations of their lives, incorporating both emotional reactions and cognitive judgments about overall life quality. As a comprehensive indicator of psychological adjustment, SWB has been consistently associated with favorable health outcomes, increased life expectancy, and stronger social connections [3]. Within this framework, life satisfaction represents the cognitive component of subjective well-being and reflects individuals' reflective appraisals of their lives as a whole. These appraisals are shaped by the perceived balance between personal aspirations and achieved outcomes and serve as an important marker of psychological well-being [4]. Furthermore, subjective well-being encompasses personal strengths and virtues that foster positive emotional states, including feelings of gratitude, fulfillment, and happiness [5].

A range of socio-demographic and occupational factors, including education, income, marital status, and employment status, has been associated with subjective well-being and life satisfaction among working women [5]. Empirical evidence suggests that women who participate in the workforce frequently report higher levels of life satisfaction compared with their non-working counterparts. Employment provides women with financial autonomy, enabling them to independently meet personal and family needs. This economic independence is often linked to enhanced feelings of security, stability, and control over living conditions. In contrast, women who are not engaged in paid work may experience reduced life satisfaction, potentially due to financial dependency and fewer opportunities for personal growth and self-realization [6].

Despite these potential benefits, employment may also introduce additional challenges for women, particularly when combined with traditional domestic responsibilities. The simultaneous management of occupational duties and household roles has been associated with increased stress, which may adversely affect subjective well-being and overall quality of life [7]. Working women often navigate competing expectations as employees, mothers, and spouses, making the reconciliation of work and family demands a persistent personal and social challenge across diverse cultural and organizational contexts [8]. The ability to effectively manage and balance work-related and non-work-related roles plays a critical role in fulfilling life responsibilities and contributes positively to overall subjective well-being [9]. So, it is crucial to explore the effect of women's work on personal well-being and satisfaction with life. To the best of the authors' knowledge, this is the first study to focus on women's work and its effects on personal well-being and life satisfaction among working and nonworking women in Egypt, in general, and in Mansoura, specifically. Thus, this study aimed to compare personal well-being (PWB) and satisfaction with life (SWL) of working vs. non-working women and their associated factors.

2. Methods

2.1. Study Design and Locality

This cross-sectional comparative study was conducted on working and non-working women who were either affiliated with or attend outpatient clinics, Mansoura campus, and Primary health care centers in Mansoura city included Talkha and Met Khamis primary health care centers, from February to April 2025.

2.2. Study Population and Sampling Method

The study population included both working and non-working in the above-mentioned settings. The inclusion criteria were being at least 18 years old, attending the outpatient clinic, or being present on the university campus as an employee, patient's companion, or visitor. In Egypt, the legal working age is 18 years; therefore, only adult women within the official working-age category were eligible for inclusion. The exclusion criteria were being under 18 years of age, pregnant, postpartum, and having chronic physical or mental diseases. A convenient sampling technique was used to recruit women.

A total of 402 women completed the questionnaire, including 210 non-working and 192 working women. While age distribution and residence did not differ significantly between working and non-working women, marital status, education, and income did. Working women were more likely to be single, have higher educational attainment, and have sufficient income (Table 1).

2.3. Sample Size Calculation

Sample size was calculated using Open Epi (https://www.openepi.com/SampleSize/). A pilot study of 35 working and 35 non-working women found that the percentages of life satisfaction were 54.3% and 40.0%, respectively. With alpha error 5%, study power of 80%, then the sample size is 186 per group. However, 402 women (192 working and 210 non-working) completed the questionnaire.

2.4. Study Tools

All study participants completed a pre-designed, interviewer-administered semi-structured Arabic questionnaire to collect the following data:

- Sociodemographic characteristics, including age, educational level, marital status, monthly income, and residence.
- Occupational history for working women, including job title, duration of employment, working hours/day, type of work shift, and working time (full-time or part-time work). Occupations were categorized according to the International Standard Classification of Occupations (ISCO-08) based on broad skill levels. Thus, high-skill occupations (skill levels 3 and 4) included managers, professionals, and associate professionals. Medium-skill occupations (skill level 2) included clerical support, technicians, service, and sales workers.

Table 1. Socio-demographic characteristics of the study groups.

Socio-demographic characteristics	Working Women N (%)	Non-Working Women N (%)	Test and P value
Age			
Mean ±SD	32.91±5.23	32.06±6.95	t=1.37, P= 0.2
≤ 35	128 (66.7)	139 (66.2)	x^2 =0.01, P=0.9
> 35	64 (33.3)	71 (33.8)	
Marital Status			
Single	49 (25.5)	12 (5.7)	x^2 =34.06, P≤ 0.001
Married	115(59.9)	174 (82.9)	
Widow or divorced	28 (14.6)	24 (11.4)	
Residence			
Urban	78 (40.6)	102 (48.6)	x^2 =2.56, P=0.1
Rural	114 (59.4)	108 (51.4)	
Education			
Secondary or Lower	63 (32.8)	160 (76.2)	x^2 =76.41, P≤ 0.001
University or Higher	129 (67.2)	50 (23.8)	
Income			
Not Enough	61 (31.8)	114 (54.3)	x^2 =20.68
Enough	131 (68.2)	96 (45.7)	P=0.004

Finally, low-skilled occupations (skill level 1) included elementary occupations and manual workers [10].

- The Life Satisfaction Scale(LSS) measures the cognitive dimension of life satisfaction and consists of 5 items. The LSS uses seven response options: Strongly Agree, Agree, Neutral, Disagree, and Strongly Disagree. The Arabic version was used [11]. Scoring was performed by summing the responses to all five items, yielding a total score ranging from 5 to 35. Higher scores indicate greater life satisfaction. Specifically, scores between 30 and 35 denote high satisfaction, 25–29 indicate satisfaction, 20–24 reflect slight satisfaction, 15–19 suggest slight dissatisfaction, 10–14 represent dissatisfaction, and 5–9 correspond to extreme dissatisfaction. The cutoff point for life satisfaction was defined as a score of 20 or higher, indicating overall life satisfaction.
- The Personal Well-being Index-Adult (PWI-A) consists of seven domains, each corresponding to a quality-of-life domain comprising standards of living, health, achievements in life, relationships, security, community-connectedness, and future security. It is measured on an 11-point end-defined Likert scale, with numerical ratings ranging from 0 (extremely dissatisfied) to 10 (extremely satisfied). An additional item, administered first, probes participants' satisfaction with their life as a whole. The question about satisfaction with religion or spirituality is optional and not included in the total score of the scale [12, 13]. The Arabic version was used [14]. Scoring was done by summing the responses to the seven core items, each rated on a 0-10 scale (0 = no satisfaction, 10 = very satisfied), and then converting the total raw score to a standard score between 0 and 100. The raw score can range from 0 to 70, which is then converted to a standard score between 0 and 100 by dividing the raw score by 7 (the number of items) and then multiplying by 10. Individual scores on the PWI can be interpreted using the following guidelines [15]: 70+ = 'normal' levels of Subjective Wellbeing, 50–69 = 'compromised' levels of Subjective Wellbeing, and 49 or less = 'challenged' levels of Subjective Wellbeing.

2.5. Data Collection and Recruitment Process

Data were collected through direct, face-to-face interviews over a period of approximately three months (February to April 2025). The interviews were conducted by two of the study authors, both with a background in public health, nursing and community medicine. Prior to data collection, they standardized the interviewing approach to ensure consistency and minimize interviewer-related bias.

Eligible women were approached in waiting areas of outpatient clinics, primary healthcare centers, and public areas within the university campus. The purpose of the study was clearly explained, and confidentiality and anonymity were assured before obtaining informed written consent.

Not all approached women agreed to participate. Some eligible women declined due to personal reasons or lack of time; however, the exact number of refusals was not recorded. A total of 402 women consented to participate and completed the questionnaire, yielding a high response rate. A pilot study was conducted with 35 working and 35 non-working women to assess the clarity and completeness of the tools and to estimate the time required to complete the questionnaire. The results showed that no refinements and modifications were needed. The pilot study was excluded from the sample.

2.6. Statistical Analysis

Data were collected, coded, computed, and analyzed using SPSS software (IBM SPSS Statistics for Windows, Version 25.0. IBM Corp., Armonk, N.Y., USA). Qualitative variables were presented as numbers and percentages. Quantitative variables were tested for normality and described as means and standard deviations (SDs). Appropriate statistical tests of significance were applied. Regression analysis was conducted to identify the independent predictors of SWL and PWB.

Predictors were organized into two main categories to identify factors associated with women's Subjective well-being and Satisfaction with life outcomes: (i) Sociodemographic predictors (e.g., age, marital status, residence, education, and income), and (ii) Work-related predictors (e.g., occupation, working hours, duration of employment, and shift work). In bivariate analyses (for each category), the Chi-square test was used to compare categorical variables, and risk ratios, along with their 95% confidence intervals, were calculated for all predictors.

Significant associations in the bivariate analyses were entered into binary stepwise logistic regression analyses (for each category) to point out the independent predictors of Subjective well-being (measured by PWI-A) and Satisfaction with life (measured by LSS). Adjusted odds ratios (AOR) and their 95% confidence intervals were calculated.

In both bivariate and logistic regression analyses, the primary outcomes—subjective well-being and life satisfaction—were treated as binary (dichotomous) variables, classifying participants into two groups: those with normal or higher well-being/satisfaction and those with compromised or lower well-being /satisfaction. This approach enabled clear interpretation of factors independently associated with a higher likelihood of subjective well-being and/or life satisfaction.

Subjective well-being (PWI-A) and satisfaction with life (LSS) were dichotomized based on validated cut-off points to facilitate clinically meaningful interpretation; therefore, binary logistic regression was applied instead of linear regression, as it provides appropriate estimation of associations (adjusted odds ratios) for categorical outcome variables.

3. Results

Table 2 illustrates that working women had significantly higher mean scores for both subjective well-being PWI-A: 72.06 ± 13.03 vs. 67.42 ± 10.14) and life satisfaction (LSS: 21.52 ± 5.76 vs. 17.55 ± 5.91) compared to non-working women. A normal level of well-being was significantly more prevalent in working women (62.0%) compared to non-working women (43.3%). Similarly, a significantly higher proportion of working women were observed in the 'satisfied' (19.3%) and 'extremely satisfied' (12.0%) categories, compared to 4.3% and 0.5% among non-working women, respectively. (Table 2)

Table 3 shows that education and income were strong independent predictors of subjective well-being. Women with university or higher education (AOR = 2.81; 95% CI: 1.83–4.31) and those reporting enough income (AOR = 2.94; 95% CI: 1.92–4.50) had better well-being outcomes.

Logistic regression analysis revealed that subjective well-being (SWB) was a significant independent

Table 2. Comparison of Satisfaction with Life (LSS) and Personal Well-being (PWI-A) scores among study groups.

Measure	Level	Working Women (n=192) N (%)	Non-Working Women (n=210) N (%)	Test and P value
Subjective Wellbeing Index (PWI-A)	Mean ±SD	72.06±13.03	67.42±10.14	t=4.03, P≤ 0.001
	Challenged Level	7 (3.6)	4 (1.9)	x^2 =17.05, P≤ 0.001
	Compromised Level	66 (34.4)	115 (54.8)	
	Normal Level	119 (62)	91 (43.3)	
Life Satisfaction Scale (LSS)	Mean ±SD	21.52±5.76	17.55±5.91	t=6.81, P≤ 0.001
	Extremely Dissatisfied	1 (0.5)	24 (11.4)	x^2 =68.55, P≤ 0.001
	Dissatisfied	24 (12.5)	32 (15.2)	
	Slightly Dissatisfied	55 (28.6)	55 (26.2)	
	Slightly Satisfied	52 (27.1)	89 (42.4)	
	Satisfied	37 (19.3)	9 (4.3)	
	Extremely Satisfied	23 (12.0)	1 (0.5)	

Table 3. Bivariate and multivariate logistic regression analysis of predictors of Subjective well-being among the study groups (n=402).

	Total	Subjective Wellbeing	Bivariate Analysis		Logistic Regression	
Predictors	**n**	**n (%) [a]**	**Risk Ratio (95%CI)**	**p**	**AOR* (95%CI)**	**P**
Overall	402	210 (52.2)	-	-	-	-
Group					-	-
Working	192	119 (62.0)	1.43 (1.18-1.73)	≤0.001		
Un-working	210	91 (43.3)	r			
Age					-	-
≤ 35	267	136 (50.9)	0.92 (0.77-1.13)	0.46		
> 35	135	74 (54.8)	r			
Residence					-	-
Rural	222	122 (55)	1.12 (0.93-1.36)	0.23		
Urban	180	88 (48.9)	r			
Education					**r**	
Secondary or Lower	223	88 (39.5)	r			
University or higher	179	122 (68.2)	1.73 (1.43-2.09)	≤ 0.001	**2.81 (1.83-4.31)**	**≤ 0.001**
Income						
Not Enough	175	62 (35.4)	r		**r**	
Enough	227	148 (65.2)	1.84 (1.48-2.30)	≤ 0.001	**2.94 (1.92-4.50)**	**≤ 0.001**
Marital Status						
Unmarried	113	57 (50.4)	1.05 (0.84-1.30)	0.65	-	-
Married	289	153 (52.9)	r			

*[a] Row percentage, AOR=Adjusted Odds Ratio, n= Number, r= Reference *Adjusted for age, residence, and marital status.*

predictor of life satisfaction. Women with higher levels of SWB were more likely to be satisfied their lives compared to those with lower SWB (AOR = 21.67; 95% CI: 12.97-36.19) (Table 4)-

Some occupational factors increased the likelihood of higher subjective well-being and satisfaction with life among working women including type of occupation, working hours, and shift work. Women employed in administrative/ technical occupations (AOR = 7.40; 95% CI: 2.69–20.35) or those in professional roles (AOR = 4.76; 95% CI: 1.81–12.55) had significantly higher levels of subjective well-being compared to unskilled and/or skilled manual workers. Conversely, shift work was a significant occupational predictor of lower well-being (AOR = 6.53; 95% CI: 2.83–15.06). Furthermore, working hours per week strongly predicted life satisfaction. Women working 40 hours or less per week were significantly more likely to report higher life satisfaction (AOR = 14.34; 95% CI: 7.09–29.02) compared to those working longer hours. (Table 5)

4. Discussion

The present study highlights the significant differences in subjective well-being and life satisfaction between working and non-working women. It contributes to the literature by exploring how socio-demographic and occupational factors are associated with the well-being of Egyptian women. Our findings indicate that it is not work alone, but also the socio-economic conditions associated with it, that play the most important role in shaping well-being outcomes.

Furthermore, we highlight the influence of occupational conditions—particularly job type, working

Table 4. Bivariate and multivariate logistic regression analysis of predictors of Satisfaction with life among the study groups (n=402).

	Total	**Satisfaction with Life**	**Bivariate Analysis**		**Logistic Regression**	
Predictors	**n**	**n (%) a**	**Risk Ratio (95%CI)**	**P**	**AOR* (95%CI)**	**P**
Overall	402	211 (52.5)	-	-	-	-
Group					-	
Working	192	112 (58.3)	1.24 (1.03-1.49)	0.02		-
Un-working	210	99 (47.1)	r			
Age						
≤ 35	267	143 (53.6)	1.06 (0.87-1.3)	0.55	-	-
> 35	135	68 (50.4)	r			
Residence						
Rural	222	123 (55.4)	1.13 (0.94-1.37)	0.19	-	-
Urban	180	88 (48.9)	r			
Education						
Secondary or Lower	223	104 (46.6)	r		-	-
University or Higher	179	107 (59.8)	1.28 (1.07-1.54)	0.009		
Income						
Not Enough	175	73 (41.7)	r	≤ 0.001	-	-
Enough	227	138 (60.8)	1.46 (1.19-1.79)			
Marital Status						
Unmarried	113	62 (54.9)	1.06 (0.87-1.30)	0.55	-	-
Married	289	149 (51.6)	r			
Subjective Wellbeing						
Normal	210	175 (83.3)	4.44 (3.29- 6.00)	≤ 0.001	**21.67 (12.97-36.19)**	**≤ 0.001**
Abnormal	192	36 (18.8)	r		**r**	

*a Row percentage, AOR=Adjusted Odds Ratio, n= Number, r= Reference; *Adjusted for age, residence, and marital status.*

hours, and shift work—on the well-being of working women. These results highlight the importance of not only enhancing women's access to work but also improving the quality of work environments to support sustainable well-being and life satisfaction.

In the current study, working women reported higher levels of subjective well-being and life satisfaction compared to non-working women (Table 2), which may be explained by the more favorable socio demographic characteristics of working women, including higher education levels and better income status (Table 1) which was similarly reported by other research [9, 16]. These findings suggest that employment provides not only financial stability but also social and psychological benefits that enhance overall wellbeing.

Several studies support the view that employment has positive effects on the mental health of women, primarily through the benefits of job engagement and economic independence. The sense of self-worth, achievement, financial stability, and social recognition associated with employment enables working women to perceive their jobs as rewarding and psychologically fulfilling [17–19]. Moreover, being confined to home has been identified as an important factor contributing to poor mental health among non-working women [20].

On the contrary, some research have reported comparable levels of life satisfaction between working and non-working women, though the underlying determinants differ. Evidence from India indicates that despite the financial independence,

Table 5. Bivariate and multivariate logistic regression analysis of occupational predictors of Subjective well-being and Satisfaction with life among working women (n=192).

	Total	Subjective Wellbeing	Bivariate Analysis		Logistic Regression	
Occupational Predictors	**n**	**n (%)** [a]	**Risk Ratio (95%CI)**	**P**	**AOR (95%CI)**	**P**
Overall	192	119 (62)	-	-	-	-
Occupation [b]						
Unskilled/skilled manual	29	8 (27.6)	r		**r**	**≤ 0.001**
Administrative/technician	75	55 (73.3)	2.66 (1.45-4.87)	≤ 0.001	**7.40 (2.69-20.35)**	**0.002**
Professional	88	56 (63.6)	2.31 (1.25-4.25)	≤ 0.001	**4.76 (1.81-12.55)**	
Working Hours/Week						
≤ 40	106	73 (68.9)	1.29 (1.02-1.63)	0.03	-	-
>40	86	46 (53.5)	r			
Duration of Employment						
≤ 10	135	79 (58.5)	r	0.12	-	-
>10	57	40 (70.2)	1.20 (0.96-1.50)			
Shift Work						
Yes	37	10 (27)	r		**r**	
No	155	109 (70.3)	2.60 (1.52-4.46)	≤ 0.001	**6.53 (2.83-15.06)**	**≤ 0.001**

	Total	Satisfaction with Life	Bivariate Analysis		Logistic Regression	
Occupational Predictors	**n**	**n (%)** [a]	**Risk Ratio (95%CI)**	**P**	**AOR (95%CI)**	**P**
Overall	192	112 (58.3)	-	-	-	-
Occupation [b]			r			
Unskilled/skilled manual	29	11 (37.9)				
Administrative/technician	75	55 (73.3)	1.93 (1.19-3.14)	≤ 0.001	-	-
Professional	88	46 (52.3)	1.38 (0.83-2.29)	0.18	-	-
Working Hours/Week						
≤ 40	106	89 (84)	3.14 (2.19-4.50)	≤ 0.001	**14.34 (7.09-29.02)**	**≤ 0.001**
>40	86	23 (26.7)	r		**r**	
Duration of Employment						
≤ 10	135	78 (57.8)	r			
>10	57	34 (59.6)	1.032 (0.80-1.34)	0.8	-	-
Shift Work						
Yes	37	21 (56.8)	r			
No	155	91 (58.7)	1.03 (0.76-1.41)	0.8	-	-

[a] Row percentage, [b] occupations classified according to the International Standard Classification of Occupations (ISCO–08), AOR=Adjusted Odds Ratio, n= Number, r= Reference

autonomy, and sense of achievement associated with employment, working women often experience greater stress due to the demands of balancing multiple roles [21]. Conversely, non-working women often find fulfillment through family roles, greater involvement in childcare, and opportunities for socialization and personal development [22]. Consequently, despite differing sources of satisfaction, both groups tend to report similar overall levels of life satisfaction.

The current study revealed that subjective well-being (SWB) was a strong independent predictor of life satisfaction (Table 4), with women with higher SWB substantially more likely to report being satisfied with their lives than those with lower SWB (AOR = 21.67). This finding aligns with previous research demonstrating that higher SWB is consistently associated with improved health outcomes, enhanced productivity, stronger interpersonal relationships, and greater overall life satisfaction [23, 24].

A range of socio-demographic and occupational factors were significantly associated with women's well-being and life satisfaction, among which employment status, education, and income play particularly important roles. In the current study, education and income consistently emerged as strong independent predictors of subjective well-being (Table 3). Women with university or higher education were nearly three times as likely to report good well-being as those with secondary or lower education (AOR = 2.81). Similarly, women who reported having enough income had almost threefold higher odds of good well-being (AOR = 2.94). These results emphasize the critical role of socio-economic empowerment in improving women's quality of life and psychological wellbeing. Similar findings have been consistently reported in earlier research linking higher education and adequate income to enhanced well-being and life satisfaction [16, 25-27].

However, previous research has shown some ambiguity regarding the relationship between education and wellbeing. Some studies have argued that educational attainment does not have a significant impact on well-being [28, 29], while others report a positive association between education and individual well-being [30]. Several researchers argue that education by itself does not directly influence happiness but that higher educational attainment enhances income, which in turn improves happiness through access to desired or scarce resources [31]. Despite these conflicting perspectives, a growing body of evidence indicates that individuals with higher educational achievements tend to report greater levels of subjective well-being [26, 32].

In this study, occupational characteristics such as job type, working hours, and shift work were significantly associated with well-being among employed women (Table 5). The current results demonstrate that occupation type is a strong predictor of subjective well-being among working women, even after adjusting for socio-demographic factors. Women in administrative or technical occupations were more than seven times as likely to report good well-being compared to unskilled and/or skilled manual workers (AOR = 7.40), while those in professional roles had nearly fivefold higher odds of well-being (AOR = 4.76). The higher well-being observed among professional and administrative/technical workers may be attributed to the psychosocial and structural advantages inherent in higher-status occupations—such as greater job autonomy, cognitive engagement, decision-making authority, and opportunities for skill development. Moreover, the professional and technical roles typically provide more stable income, access to benefits, and social recognition, all of which are important determinants. These findings emphasize that occupational class and job quality are key determinants of mental health and wellbeing, especially among women balancing multiple roles [25, 33].

In contrast, women in manual or lower-skilled occupations frequently encounter job insecurity, low wages, limited advancement opportunities, and exposure to unsafe or physically demanding environments, contributing to chronic stress and diminished satisfaction [34]. Additionally, gender-based occupational segregation may exacerbate these disparities, as women in manual sectors frequently encounter limited control, lower social support, and fewer opportunities for career progression [35].

On the contrary, shift work was associated with lower well-being; women who did not work shifts were more than 6 times as likely to experience good well-being (AOR = 6.53). This aligns with evidence linking shift work to sleep deprivation, circadian rhythm disruption, and poorer physical and psychological health [36]. Sleep deprivation is known to impair cognitive and emotional functioning, thereby diminishing overall well-being [37].

Furthermore, weekly working hours were strongly associated with higher life satisfaction. Women working ≤40 hours/week were significantly more likely to report higher life satisfaction, with odds approximately 14 times greater than those working longer hours (AOR = 14.34). This finding supports previous research suggesting that shorter working hours promote life satisfaction by allowing better work–life balance [38]. International literature similarly highlights that satisfaction is driven by factors beyond income, including working hours, work–life balance, and social support [39]. Cross-cultural evidence indicates that women's preferences for work hours are influenced by sociocultural norms and role expectations. In many settings, women prefer shorter workweeks due to dual roles in employment and caregiving, whereas, in high-income contexts such as the United States, longer hours are often perceived as a means of achieving economic security and professional advancement, which may be associated with greater subjective well-being and job satisfaction [40].

The interpretation of these findings should be considered within the socio-cultural context of Egypt, where women's roles are shaped by both traditional expectations and evolving economic demands. In settings such as Mansoura, employment may be associated not only with financial benefits but also with greater social recognition and autonomy, which may, in turn, be linked to higher perceived well-being. At the same time, many working women continue to bear primary responsibility for household and caregiving roles, which may contribute to role strain. Conversely, non-working women may derive satisfaction from family roles and social cohesion, but may also experience financial dependency or fewer opportunities for personal development. These contextual factors may partly explain the observed differences between working and non-working women and underscore the importance of interpreting the findings within the local cultural framework.

5. Conclusion and Recommendation

This study demonstrates that employment is positively associated with women's subjective well-being and life satisfaction, primarily through higher education, adequate income, and improved socio-economic status. Beyond financial benefits, employment enhances women's sense of self-worth, autonomy, and social recognition. However, the nature of work significantly shapes these outcomes: women in professional or technical occupations reported greater well-being than those in manual or shift-based jobs, reflecting the importance of job autonomy, security, and supportive work environments. Shorter working hours were also linked to higher life satisfaction, emphasizing the need for balance between professional and family roles. Policymakers and employers should prioritize interventions that enhance job quality, ensure reasonable working hours, and provide job security. Future research should adopt longitudinal designs to confirm causal relationships and explore the long-term effects of employment characteristics and changing work conditions on women's wellbeing.

5.1. Strengths and Limitations

This study possesses several notable strengths. It addresses a research gap by comparing the subjective well-being and life satisfaction of working and non-working women in Egypt—particularly in the understudied context of Mansoura. The use of validated Arabic versions of internationally recognized instruments, namely the Life Satisfaction Scale(LSS) and the Personal Well-being Index–Adult (PWI-A), ensures high reliability and comparability of results. Furthermore, the analysis incorporated a wide range of socio-demographic and occupational variables, offering a comprehensive understanding of the multifactorial determinants of women's wellbeing. The relatively large and statistically well-powered sample further enhances the credibility and generalizability of the findings. Importantly, the study provides practical insights for policymakers, public health professionals, and workplace administrators seeking to improve women's quality of life and mental health through evidence-based interventions.

Despite its strengths, the authors acknowledge that this study has limitations. The cross-sectional design restricts the ability to establish causality between employment-related factors and well-being outcomes. Additionally, the use of a convenience sampling technique may limit the generalizability of the findings to all Egyptian women, as participants were recruited from specific community and institutional settings within Mansoura. Furthermore, significant baseline differences between working and non-working women—particularly in education, income, and marital status—may have influenced the observed associations, as these factors are closely linked to employment status. Although multivariate regression was used to adjust for these potential confounders, residual confounding cannot be excluded. In addition, some adjusted odds ratios were associated with relatively wide confidence intervals, which may reflect small subgroup sizes after dichotomization of outcomes. Therefore, these findings should be interpreted with caution.. The reliance on self-reported data may introduce reporting or recall bias, as responses reflect participants' subjective perceptions rather than objective measures. The assessment of chronic physical and mental illnesses relied on self-report, which may have led to underreporting, particularly for mental health conditions due to stigma or lack of formal diagnosis. Additionally, some participants were recruited as patients' companions or visitors, and their responses may have been influenced by temporary stress related to their companion's health condition, which could have affected subjective well-being and life satisfaction measures. Moreover, the study was geographically confined to Mansoura city and may not capture variations in employment conditions or cultural contexts across other regions of Egypt. Finally, certain potential confounders—such as personality traits, family support, and broader cultural influences—were not assessed. Additionally, some factors that may influence women's well-being—such as the quality of marital relationships, the number of children, the availability of childcare support, and household responsibilities—were not assessed, which may have affected the observed associations.

Institutional Review Board Statement: The study protocol was approved by the Institutional Review Board (IRB), Faculty of Medicine, Mansoura University, (code number: R.24.12.2926).

Informed Consent Statement: Following the assurance of confidentiality and anonymity of data, which are never to be used for any other purpose than scientific research, informed written consent was obtained from participants who agreed to share in the study.

Acknowledgments: The authors thank all study participants who generously agreed to participate

Declaration of Interest: The authors declare no conflict of interest

Author Contribution Statement: HM, AA, AE, SA and EE were responsible for the initial conception and design of the study. HM and EE collected data. AA ran statistical analysis and participated in the interpretation of the data. AE revised the data analysis and interpretation. AA and SA drafted the initial version of the manuscript and all authors were involved in critically revising the manuscript. All authors have reviewed and approved the final version of the manuscript.

References

1. Parashar M, Singh M, Lal P. Life satisfaction and correlates among working women of a tertiary care health sector: A cross-sectional study from Delhi, India. *Med J Dr DY Patil Univ*. 2018;11(5):406-411. Doi: 10.4103/mjdrdypu.MJDRDYPU_240_17
2. Ruggeri K, Garcia-Garzon E, Maguire Á, Matz S, Huppert F. Well-being is more than happiness and life satisfaction: a multidimensional analysis of 21 countries. *Health Qual Life Outcomes*. 2020;18:192. Doi: https://doi.org/10.1186/s12955-020-01423-y
3. Das KV, Jones-Harrell C, Fan Y, et al. Understanding subjective well-being: perspectives from psychology and public health. *Public Health Rev*. 2020;41:25. Doi: 10.1186/s40985-020-00142-5.
4. Diener E, Emmons RA, Larsen R, Griffin S. The Satisfaction With Life Scale. *J Pers Assess*. 1985;49(1):71 75. Doi: https://doi.org/10.1207/s15327752jpa4901_13
5. Cintantya D, Nurtjahjanti H. Hubungan antara work-life balance dengan subjective well-being pada sopir taksi PT. *J Empati*. 2020;7(1):339-344. Doi: https://doi.org/10.14710/empati.2018.20246
6. Machhi YV, Thakor ID. Life satisfaction among working and non working women. *Int J Indian Psychol*. 2024;12(1):1676-1682. Doi: 10.25215/1201.155.
7. Kuykendall L, Tay L, Ng V. Leisure engagement and subjective well-being: A meta-analysis. *Psychol Bull*. 2015;141(2):364. Doi: 10.1037/a0038508
8. Elegbede C, Abidogun M. Women's experience of balancing work and family roles: counselling strategies for

promoting work life balance. 10th International Congress on Humanities and Social Science in a Changing World; 2023:1 10. Doi: 10.6084/m9.figshare.24041868.
9. Sintha D, Adnans A. The influence of work life balance to subjective well-being on women working in PTPN II. *Int J Res Rev*. 2024;11(7):211-220. Doi: https://doi.org/10.52403/ijrr.20240723
10. International Labour Organization (ILO). International Standard Classification of Occupations: ISCO-08. Volume I. Structure, Group Definitions and Correspondence Tables. 2012. Available online: https://www.ilo.org/publications/international-standard-classification-occupations-2008-isco-08-structure (Last Accessed on 15 September 2025).
11. Zawawi J, Hamaideh S. Satisfaction With Life Scale—Arabic Version (LSS). *APA PsycTests*. 2009. Doi: https://doi.org/10.1037/t63505-000
12. International Well-being Group. Personal Well-being Index. 5th ed. Melbourne: Australian Centre on Quality of Life, Deakin University. 2013. Available online: https://novopsych.com/assessments/well-being/personal-wellbeing-index-adult-5-pwi-a/ (Last Accessed on 12 May 2025).
13. Khor S, Cummins RA, Fuller-Tyszkiewicz M, et al. Australian Unity Well-being Index: – Report 36: Social connectedness and wellbeing. Geelong: Australian Centre on Quality of Life, School of Psychology, Deakin University. 2020. Available online: http://www.acqol.com.au/publications#Open-access. (Last Accessed on 12 May 2025).
14. Lotfy NA. Validity and reliability of the Arabic version of the” Personal Well-being Index-Adults” on adults with hearing impairment. *Health Promot Perspect*. 2020;10(3):250-256. Doi: 10.34172/hpp.2020.39
15. Tomyn AJ, Weinberg MK, Cummins RA. Intervention efficacy among ‘at risk’adolescents: a test of subjective well-being homeostasis theory. *Soc Indic Res*. 2015;120(3): 883-895. Doi: 10.1007/s11205-014-0619-5
16. Sinha S. Multiple roles of working women and psychological well-being. *Ind Psychiatry J*. 2017;26(2):171-177. Doi: 10.4103/ipj.ipj_70_16
17. Jalil MF, Ali A. The influence of meaningful work on the mental health of SME employees in the COVID-19 era: can coping strategies mediate the relationship?. *BMC Public Health*. 2023;23:2435. Doi: 10.1186/s12889-023-17347-3
18. Kuppusamy J, Angusamy A, Anantharaman RN. Life Satisfaction among working women in a Southeast Asian country. *Rev Appl Socio Econ Res*. 2023;25(1):125-137. Doi: https://doi.org/10.54609/reaser.v25i1.209
19. Lu Z, Yan S, Jones J, He Y, She Q. From housewives to employees, the mental benefits of employment across women with different gender role attitudes and parenthood status. *Int J Environ Res Public Health*. 2023;20(5):4364. Doi: https://doi.org/10.3390/ijerph20054364
20. Zhang Y, Gao Y, Zhan C, Liu T, Li X. Subjective Well-Being of Professional Females: A Case Study of Dalian High-Tech Industrial Zone. *Front Psychol*. 2022;13:904298. Doi: 10.3389/fpsyg.2022.904298
21. Agnes M, Akhila PJ. Mental Health and Life Satisfaction: A Comparative Study Among Working and Non-Working Women. *Int J Indian Psychol*. 2023;11(2). Doi: https://doi.org/10.25215/1102.230
22. Sharma V, Rana P, Tripathy M. A Comparative Study of Life Satisfaction between Working and Non-Working Women in A Spiritual Environment. *Int J Arts Humanities Soc Stud*. 2023;5(3):52-54. Doi: 10.5281/zenodo.8026610
23. Boehm JK, Peterson C, Kivimaki M, Kubzansky L. A prospective study of positive psychological well-being and coronary heart disease. *Health Psychol*. 2011;30(3):259. Doi: https://doi.org/10.1037/a0023124
24. Almeida PM, Ramos PA. The Subjectivity of Subjective Well-Being. *SAGE Open*. 2025;15(1). Doi: 10.1177/21582440251329860
25. Diener E, Lucas RE, Oishi S. Advances and Open Questions in the Science of Subjective Well-Being. *Collabra Psychol*. 2018;4(1):15. Doi: 10.1525/collabra.115
26. Tran DB, Pham TDN, Nguyen TT. The influence of education on women’s well-being: Evidence from Australia. *PLoS One*. 2021;16(3):e0247765. Doi: 10.1371/journal.pone.0247765
27. Özer U, Fidrmuc J. Education, income, and subjective wellbeing: Cross-country evidence. *J Happiness Stud*. 2025;26(1):45–67. Doi: https://doi.org/10.21547/jss.1552896
28. Duke N, Macmillan R. Schooling, skills, and self rated health: A test of conventional wisdom on the relationship between educational attainment and health. *Sociol Educ*. 2016;89(3):171 206. Doi: 10.1177/0038040716653168.
29. Lee KS, Yang Y. Educational attainment and emotional well-being in adolescence and adulthood. *SSM Ment Health*. 2022;2:100138. Doi: 10.1016/j.ssmmh.2022.100138
30. Salinas-Jiménez MDM, Artés J, Salinas-Jiménez J. How do educational attainment and occupational and wage-earner statuses affect life satisfaction? A gender perspective study. *J Happiness Stud*. 2013;14(2):367-388. Doi: 10.1007/s10902-012-9334-6
31. Jiang YL. The path analysis of the effect of education on subjective well-being. *Adv Psychol*. 2019;9(4):726-738. Doi: 10.12677/AP.2019.94090.
32. Nikolaev B. Does other people’s education make us less happy?. *Econ Educ Rev*. 2016;52:176-191. Doi: 10.1016/j.econedurev.2016.02.005.
33. Clausen T, Pedersen LRM, Andersen MF, Theorell T, Madsen IEH. Job autonomy and psychological well-being: A linear or a non-linear association? *Eur J Work Organ Psychol*. 2021;31(3):395–405. Doi: 10.1080/1359432X.2021.1972973

34. Biswas A, Harbin S, Irvin E, et al. Differences between men and women in their risk of work injury and disability: A systematic review. *Am J Ind Med*. 2022;65(7): 576–588. Doi: 10.1002/ajim.23364.
35. Sahni S, Kaushal LA, Gupta P. Gendered differences and strategies for work-life balance: Systematic review based on social ecological framework perspective. *Acta Psychol*. 2025;256:105019. Doi: 10.1016/j.actpsy.2025.105019.
36. Lee HY, Bjorvatn B. Shift work, sleep, and wellbeing: Evidence from longitudinal studies. *J Occup Health Psychol*. 2024;29(2):123-135. Doi: 10.2486/indhealth .2024-0088.
37. Ramar K, Malhotra RK, Carden KA, et al. Sleep is essential to health: an American Academy of Sleep Medicine position statement. *J Clin Sleep Med*. 2021;17(10): 2115-2119. Doi: 10.5664/jcsm.9476.
38. Shao Q. Exploring the promoting effect of working time reduction on life satisfaction using Germany as a case study. *Hum Soc Sci Commun*. 2022;9(1):1-8. Doi: 10.1057/s41599-022-01480-2.
39. Gorry A, Gorry D, Slavov SN. Does retirement improve health and life satisfaction?. *Health Econ*. 2018;27(12):2067-2086. Doi: 10.1002/hec.3821.
40. Lombardo P, Jones W, Wang L, Shen X, Goldner EM. The fundamental association between mental health and life satisfaction: results from successive waves of a Canadian national survey. *BMC Public Health*. 2018;18(1):342. Doi: 10.1186/s12889 018 5235 x.